



Shree Swaminarayan College of Computer Science

BCA-604 core java

Assignment -3

Single inheritance

assignment 1,2 submission date: 17,18 / 08 / 2026

- (1) Write a program in java to one base class and one derived class named costprice(base) and profitprice (derived) follow below instructions:
 - a. costprice clas take user input: product id, product name, productcost
 - b. profitprice class: calculate and display of product profit in derived class

Multilevel Inheritance

- (2) Write a program in java to three class name gold, silver, platinum and follow below instructions:
 - a. gold class: take user input gold weight, gold price
 - b. silver class: take user input silver weight, silver price
 - c. platinum class: user input platinum weight, platinum price
 - d. and display message according to their price like(platinum>gold>silver)
- (3) Write a java program to three class called history,ancient,modern and follow below instructions
 - a. historyclass: enter id for user ,enter time like 200BC or 300AD
 - b. ancient: before christ (B.C)
 - c. modern: anno domini (A.D)
 - d. if user enter 200BD then show message **ancestors** and if user enter 300AD then show message **descendants** and use proper method.

Hierchical inheritance

- (4) Write a program in java to create three clas name **zoo**(main class),animalsection, birdsection and follow below instructions
 - a. Animalsection: take user input enclosureid, enclosurename, animal name
 - b. Birdsection: take user input enclosureid, enclosurename, animal name
 - c. Displayall information using appropriate method
- (5) Write a program in java to create class called fruit(Main clas),citrus, sweet and follow below instructions
 - a. fruit: take user input like fruitname, fruit price
 - b. citrus: take user input citrus fruit name ,fruit price
 - c. citrus: take user input citrus fruit name ,fruit price
 - d. All name display properly and use proper method.
- (6) Write a program to create three class called heaven(main),god, angel and follow below instructions
 - a. Heaven class: take user input for take door number to god like 1 and 2
 - b. God:take user input for take user input like godgate
 - c. Angel: take user input for take user input like angelgate
 - d. Displayall data in proper format and use of suer keyword and method



Method overriding

- (7) Write a program in java to calculate probability of getting a Heart of queen from 52 deck of card using method overriding use class like
- card(main): enter user id, user name
 - probability: check probability
 - And print all data and also use proper method

(Hint: probability=possible outcome/total outcome)

Abstract class:

(8) Write a program in java to create abstract class called mall and abstract method called product and take dynamically user input from user like item id, item price item quantity and print total price and use proper method to display data.

(9) Write a program in java to perform all basic mathematical operation using abstract class and abstract method

Interface

Using multiple inheritance

(10) Java Program to Implement Shape Interface to calculate area of circle , rightangle triangle, square using interface.

Using extend method only

(11) Java program to find square of any two digit number using shortcut method and interface
(Hint: $(a+b)^2 = a^2 + 2*(a)*(b) + (b)^2$)

Using Hybrid inheritance

(12) write a program to ceate interface country, india, japan, village and print message like
Country base interface

India extend or implements from country

japan extends or implements from country

Village extends or implements from india,japan

Show message by user input : india: country of god, japan: country of paradise, village: village of god and paradise

Area and volume

(13) Write a program in java to find square and rectangle area using single level inheritance

(14) perform a program to find area of all triangle using multilevel inheritance(Triangle,right triangle,equilateral triangle,isosceles triangle)

(15) Perform a program in java to find circle,semi circle, Ring(shaded region),sector of a circle using hierchical inheritance

(16) perform program to find parallelogram,Rhombus,Trapezium Area use multilevel inheritance.

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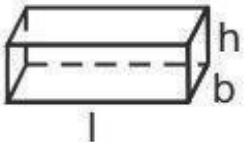
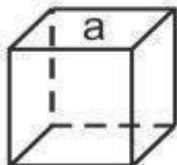
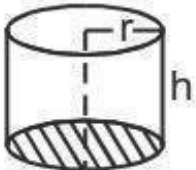
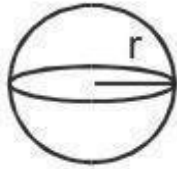
Assignment -3

Name	Figure	Perimeter	Area
Rectangle		$2(a + b)$	ab
Square		$4a$	a^2
Triangle		$a + b + c = 2s$	$1 = \frac{1}{2} \times b \times h$ $2 = \sqrt{s(s-a)(s-b)(s-c)}$
Right triangle		$b + h + d$	$\frac{1}{2} bh$
Equilateral triangle		$3a$	1. $\frac{1}{2} ah$ 2. $\frac{\sqrt{3}}{4} a^2$
Isosceles right triangle		$2a + d$	$\frac{1}{2} a^2$
Parallelogram		$2(a + b)$	ah
Rhombus		$4a$	$\frac{1}{2} d_1 d_2$
Trapezium		Sum of its four sides	$\frac{1}{2} h(a + b)$
Circle		$2\pi r$	πr^2
Semicircle		$\pi r + 2r$	$\frac{1}{2} \pi r^2$
Ring (shaded region)		----	$\pi(R^2 - r^2)$
Sector of a circle		$l + 2r$ where $l = \frac{\theta}{360} \times 2\pi r$	$\frac{\theta}{360} \times \pi r^2$

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Name	Figure	Curved Surface area	Total surface area	Volume
Cuboid		$2h(l + b)$	$2(lb + bh + lh)$	lbh
Cube		$4a^2$	$6a^2$	a^3
Right circular cylinder		$2\pi rh$	$2\pi r(r + h)$	$\pi r^2 h$
Right circular cone		πrl	$\pi r(l + r)$	$\frac{1}{3}\pi r^2 h$
Sphere		—	$4\pi r^2$	$\left(\frac{4}{3}\right)\pi r^3$
Hemi-sphere		$2\pi r^2$	$3\pi r^2$	$\left(\frac{2}{3}\right)\pi r^3$

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BCA-604 core java

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Package:

(1) Write a program to create package name maths and write first class basicinfo and perform basic mathematical operation on it like(+,-,*,/) and write second file class quant and display all information

(2) Write a program to create package name convert and first class compute and perform operation like

(1) meter to centimeter (1 m = 100 cm)

(2) fahrenheit to celsius ($c = (f - 32) / 1.8$)

(3) rim to paper (1 rim = 500 paper)

And second class like cmain and display all data and input taken by user and dynamically Perform operation

(3) Write a program to create package news and first class file is newspaper and perform operation Like take user input and

(1) newspaper name

(2) newspaper price

(3) newspaper language (English / Gujarati)

and second class like newsinfo and display all data and input taken by user and dynamically perform operation

(4) Write a program to create package volume and first class file is geometry and take user input dynamically and perform operation.

(1) cube volume

(2) cuboid volume

(3) Right circular cylinder volume

(4) Right circular cone volume

(5) sphere volume

(6) hemi sphere volume

And second class like display volume and display all data